

Wearable-Based Behavioural Monitoring for Dementia Care: From Dataset Development to Routine-Aware Anomaly Detection

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Abstract

INTRODUCTION: Disruption of daily routines, such as irregular sleep, skipped meals or reduced mobility, is often among the earliest observable markers of functional decline in dementia, yet continuous, unobtrusive monitoring of this disruption remains difficult to build and validate at scale, mainly due to the scarcity of realistic wearable datasets and of longitudinal behavioural data.

OBJECTIVES: This paper reports two complementary contributions developed within the same wearable-monitoring project: the incremental construction of a real-world multimodal wearable dataset for human activity recognition, and a routine-aware behavioural monitoring framework that uses that dataset to train and evaluate a sequential anomaly detection model.

METHODS: A dual-device wearable setup, combining a wrist-worn inertial prototype with an external spot-check physiological monitor, was used to collect labelled activities of daily living from twelve adult participants under supervised conditions. The resulting activity database seeds a knowledge-informed synthetic daily-routine generator and trains a Bidirectional Gated Recurrent Unit (BiGRU) sequential model for context-dependent anomaly detection, evaluated on synthetic routines with injected behavioural deviations.

RESULTS: Activity-recognition baselines trained on the dataset reach an accuracy of 0.73 (LightGBM), while the anomaly detection model attains an AUROC of 0.999 and a PR-AUC of 0.989 on synthetic routines with injected behavioural deviations.

CONCLUSION: These results support the feasibility of the acquisition pipeline and of the routine-modelling approach as a foundation for future validation with longitudinal, real-world dementia cohorts.

Keywords: Wearable Sensors; Human Activity Recognition; Dataset Development; Behavioural Anomaly Detection; Synthetic Data Generation.

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1. Introduction

Dementia has become one of the defining public health challenges of our time. The World Health Organization estimates that around 55 million people currently live with dementia worldwide, with close to 10 million new cases diagnosed every year [1], and Alzheimer's Disease International reports this as roughly one new diagnosis every three seconds globally [2]. Rather than a single disease, dementia is an umbrella term for a group of neurodegenerative conditions; including Alzheimer's disease, vascular dementia, Lewy body dementia and frontotemporal dementia, that progressively erode memory, reasoning, communication and behavioural control, ultimately compromising an individual's independence in everyday life.

Clinically, the way this decline manifests day to day is at least as informative as any single test result. Observational work in long-term care shows that people living with dementia tend to organise their day around a small number of anchor activities such as meals, hygiene, and rest and that disruption of these anchors, through boredom, disorganized routines or reduced engagement, tracks closely with functional and cognitive status [3]. When these anchors shift or disappear, whether through a skipped meal, a disrupted sleep–wake cycle, or an unexplained drop in mobility, the change is frequently a marker of worsening status, and a predictor of falls, malnutrition or hospitalisation. The burden of tracking these changes falls overwhelmingly on family caregivers, who report high rates of depression, anxiety and burnout, particularly when behavioural disturbances or night-time disruptions are involved [4].

Wearable devices are increasingly proposed as a way to lighten this burden. Equipped with accelerometers, gyroscopes and physiological sensors, they can continuously capture fine-grained motion and vital-sign patterns without the infrastructure dependence, cost, or privacy trade-offs of camera-based or ambient sensing [5]. When paired with machine learning, this stream of data supports Human Activity Recognition (HAR): the automatic classification of behaviours such as walking, sitting, sleeping, or eating [6]. Activity labels alone, however, answer only part of the clinical question. Knowing that someone is sitting is less useful than knowing whether sitting at that hour, for that long, is typical for that person, which requires modelling the temporal organisation of behaviour, not just isolated instance of it.

Two practical obstacles stand in the way of building and validating such routine-aware systems. First, publicly available HAR datasets are dominated by short, laboratory-controlled recordings that do not reflect the sensing conditions, activity mix, or noise characteristics of everyday, unscripted use, and datasets collected specifically from people with dementia are even scarcer, constrained by ethical review, participant burden, and the cost of reliable long-term annotation [6]. Second, because behavioral anomalies are by definition rare and highly

individual, supervised learning has very little labelled anomalous data to learn from, and this is compounded when no longitudinal recordings exist at all.

Wearable sensing is a natural fit for capturing this structure: it is well established for location tracking, fall detection, sleep monitoring and activity tracking in dementia care [7], with physiological sensors such as heart rate monitors additionally supporting stress and sleep-quality assessment [8], and several commercial GPS watches and medical-alert devices already target this population [9], [10], [11]. Most deployed systems, however, remain limited to descriptive monitoring and threshold-based alerts, rarely interpreting behavioural structure or detecting context-dependent deviations [12].

The datasets that could support richer models are themselves a limiting factor. Benchmark datasets such as PAMAP2 [13] and Opportunity [14], and more recent releases such as the PADS smartwatch dataset [15] and openly shared clinical activity datasets [16], have driven progress in wearable-based HAR, but remain short, laboratory-style, or narrowly scoped, and dedicated datasets for older or cognitively impaired populations are scarcer still, constrained by ethical approval, recruitment and annotation cost [17]. On the modelling side, HAR has progressed from handcrafted features and classical classifiers [18], [19] to convolutional and recurrent architectures learned directly from raw signals [20], [21], supporting mobility assessment, rehabilitation and fall detection [22], [23], and, specifically for dementia, wrist accelerometry has even been explored as a predictor of incident Alzheimer's disease [24]. Across this literature, however, activity recognition is typically treated as an end in itself rather than as an input to a higher-level model of routine structure.

Modelling that higher-level structure is the goal of behavioural anomaly detection, approached with autoencoders and recurrent networks [25] and, more recently, attention-based architectures such as the Anomaly Transformer [26] and decomposition-based multi-scale transformers [27]. Within dementia care, anomaly detection has targeted wandering, disrupted sleep, missed meals and prolonged inactivity [28], [29], but typically relies on limited datasets and short observation windows. Because continuous, multi-day recordings from people living with dementia remain scarce given adherence, ethical and annotation constraints [12], synthetic data generation has become a common response across healthcare more broadly [30], [31], [32], [33], letting researchers stress-test models and target rare events without exposing identifiable data, provided generated sequences preserve physically plausible dynamics [31].

This article addresses the two obstacles identified above, the shortage of realistic HAR datasets and the shortage of longitudinal behavioural data for anomaly detection, as parts of a single pipeline developed within a broader wearable-monitoring project. It first documents the early-stage development of a multimodal wearable dataset for activities of daily living, describing the dual-device acquisition setup, the supervised collection protocol, and

the resulting data characteristics [34]. It then shows how the activity vocabulary established by that dataset feeds into a routine-aware behavioral monitoring framework, in which synthetic daily routines, generated from empirically grounded temporal patterns, are used to train and evaluate a BiGRU-based sequential model for context-dependent anomaly detection. The two strands are deliberately reported together: the dataset supplies the activity semantics that make the synthetic routines behaviourally meaningful, while the routine-modelling framework demonstrates the downstream purpose the dataset is meant to serve.

The novelty of this work lies neither in the use of wearables for dementia monitoring nor in the use of recurrent networks for anomaly detection, both of which have precedents [24], [35]–[38], but in their integration into a single, traceable pipeline in which each stage constrains the next. Existing dementia-oriented anomaly detection studies either rely on ambient smart-home sensors, often with activity labels taken from public datasets and a limited number of manually injected abnormal episodes, or analyse wearable and physiological signals for a specific symptom, such as agitation, or for long-term risk prediction, without

modelling the daily routine as a sequence, Table 1. In contrast, the present work derives its activity data from a purpose-built multimodal wearable acquisition system rather than from an existing ambient-sensor benchmark. Together with time-of-day structure and duration ranges drawn from dementia-care literature, to parameterize a knowledge-informed generator of complete 24-hour routines, rather than inserting isolated hand-crafted episodes into short recordings; and evaluates a routine-aware sequential detector that labels every 5-minute step according to its timing, duration and surrounding context.

The system is presented as a proof of concept at an early technology readiness level, not as clinically deployable technology.

The remainder of the article is organised as follows. Section 2 presents the methodology, covering the system and project context, the development of the wearable dataset, and the behavioural monitoring framework. Section 3 reports experimental results for both activity recognition and anomaly detection. Section 4 discusses findings, limitations and deployment considerations, and Section 5 concludes with directions for future work.

Table 1. Comparison of the present work with closely related studies on behavioural anomaly detection and wearable monitoring in dementia

Study	Sensing modality	Data and population	Anomaly ground truth	Method and output
[24]	Wrist accelerometry (10 days)	1,270 older adults	Real; incident Alzheimer’s disease within 5 years	Neural network; person-level risk prediction; no routine model
[29]	Wrist wearable (ACC, BVP, EDA, skin temperature)	17 inpatients with dementia; up to 8 weeks	Real, video-annotated agitation and aggression events	Personalised ML on 1-min windows; symptom detection (median AUC 0.87); no routine model
[35]	Ambient (PIR, motion, pressure, door, energy) plus twice-daily physiological readings	12 people with mild-to-moderate dementia; 6-month home trial	Real; clinical-team notifications as reference	Markov-chain entropy for unusual days; HMM-based fusion for agitation (AUC 0.87)
[36]	Ambient (van Kasteren smart-home dataset)	3 households, non-dementia; < 1 month	Synthetic: 135 manually injected abnormal segments	Vanilla RNN, LSTM and GRU vs. HMM and CRF; window-level detection
[37]	Ambient (public smart-home datasets)	Non-dementia residents	Synthetic abnormal activities	CNN; activity recognition and deviation detection
[38]	Ambient (motion and door-entry sensors)	Real homes of older adults with dementia	Real sensor data; no step-level anomaly labels	Clustering of abnormal patterns; recurrent networks forecasting sensor activity
Our study	Wrist IMU (25 Hz) plus HR/SpO₂ spot-check monitor (1 Hz)	12 healthy young adults, 13.3 h (HAR); synthetic 24-hour routines (anomaly detection)	Synthetic normal routines and synthetic anomalies	LightGBM activity recognition; BiGRU step-level, routine-aware anomaly detection

2. Methodology

The work reported here is part of a larger project developing an end-to-end wearable-based monitoring system for dementia care, Figure 1, comprising custom wrist-worn [39] and in-ear [40] devices under active development. The target architecture follows three layers: data acquisition through purpose-built wearables, data processing and behavioural modelling, and a monitoring interface for caregivers and clinicians. At the acquisition layer, the devices combine inertial sensing (accelerometer, gyroscope), physiological sensing (heart rate, body temperature) and, where available, location information from Bluetooth beacons or GPS. Sensor data is transmitted to a local base station or mobile gateway over low-power wireless links, then processed into activity labels and behavioural summaries that feed the caregiver-facing dashboard.

2.1. Wearable Dataset Development for Human Activity Recognition

This subsection documents the construction of the real-world wearable dataset used both to validate the acquisition pipeline in its own right and to supply the activity vocabulary consumed by the behavioural monitoring framework.

Data was collected from twelve healthy adult volunteers with no reported mobility impairments, aged 23–28 years

(mean 24.45, SD 1.54, median 24). Because each participant performed multiple repetitions of the target activities, the resulting sample of activity segments carries a sex distribution of 30% male and 70% female. The acquisition protocol was reviewed and approved by the hosting institution's Ethics Committee prior to data collection, consistent with standard practice for wearable studies involving human participants. All participants were informed of the study's purpose and gave consent, no personally identifiable information was stored alongside sensor data, and all recordings were anonymised during processing. The protocol did not involve vulnerable populations at this stage, its sole purpose was to validate the acquisition infrastructure and dataset structure ahead of future work with the target clinical population.

At the time the dataset was collected, the custom wearable device was still at an intermediate stage of hardware development. Data acquisition therefore relied on a dual-device setup: a prototype wrist-wearable for inertial sensing, based on the hardware architecture reported in [39], paired with an external, commercially available spot-check monitor [41] for physiological measurements. This arrangement made it possible to validate the acquisition, annotation and multimodal alignment procedures ahead of the final integrated sensing hardware, while still producing activity-level data consistent with the categories the finished system is intended to recognise: postural states, mobility-related actions, and hygiene and eating-related activities.

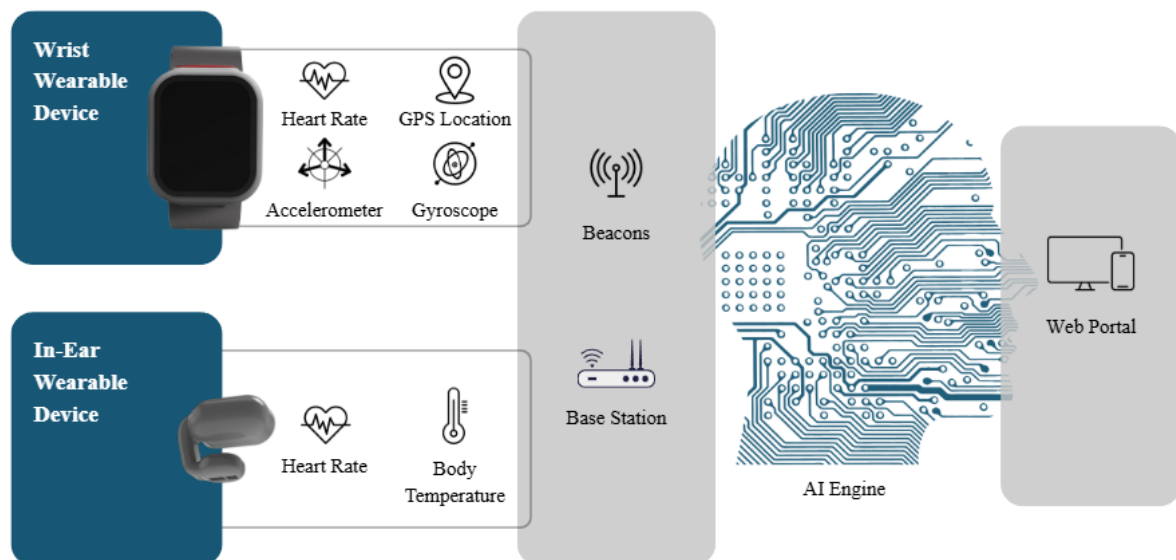


Figure 1. Overview of the wearable-based monitoring system architecture.

Inertial data was captured continuously by the wrist-wearable prototype at 25 Hz, recording tri-axial accelerometer and gyroscope signals through its BMI270 IMU, a sampling rate chosen to resolve fine-grained postural transitions and mobility dynamics. In parallel, an external PC-303 spot-check monitor recorded heart rate and peripheral oxygen saturation (SpO₂) at 1 Hz. Both devices ran over the same acquisition window for each activity, so that inertial and physiological streams, despite their different sampling rates, always covered identical time intervals.

All sessions were supervised to ensure correct device placement and consistent execution. Participants performed activities from a fixed set spanning postural states (sitting, standing, laying), mobility (walking, light movement), hygiene (washing hands, brushing teeth, showering) and eating (with cutlery, with hands), executing each one naturally rather than under strict motion constraints, and repeating activities multiple times to increase variability. An operator logged the start and end of each execution using a semi-automated timer on the wearable system, applied simultaneously to both devices, which produced precise, consistent segment boundaries for later annotation.

Because each recording session was itself bounded by an explicit start and stop command tied to a single pre-selected activity, annotation reduced to associating each resulting segment with that activity label, no post-hoc segmentation of continuous streams was required. This produced an unambiguous, one-activity-per-segment ground truth, later cross-checked against the operator's activity log for consistency.

The mismatch between the 25 Hz inertial stream and the 1 Hz physiological stream was resolved at the time-step level rather than the activity level, by forward-filling each physiological measurement across the intervening inertial samples until a new reading arrived. This preserves a physiological value at every inertial time step without introducing artificial dynamics through interpolation. Missing values at the very start of a recording were filled with the first available measurement, and any gaps within a recording were filled with the last observed value.

The dataset totals approximately 13.3 hours of recording across ten activity classes. Figure 2 summarises the relative contribution of each class to total recording time.

Postural, low-mobility states dominate: seating and laying together account for more than half of the total recording time, consistent with the prevalence of low-intensity states in everyday routines and providing ample data for posture-related recognition. Eating-related activities are the next best represented, together spanning around a fifth of the dataset and offering coverage of the upper-limb, fine-motor patterns that are clinically informative for assessing self-care and nutritional autonomy. Mobility (walking, light movement) and hygiene (washing teeth, showering, washing hands) classes occupy smaller but still meaningful shares, each exhibiting

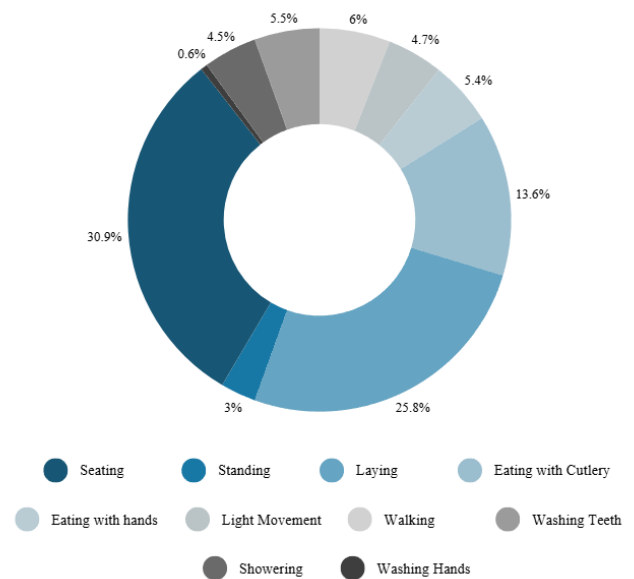


Figure 2. Distribution of recorded activities as a percentage of total recording time

distinctive motion signatures despite their shorter typical duration.

This distribution should be read as validating the acquisition, annotation and processing pipeline rather than as a representative sample of long-term daily behaviour classes such as sleeping and medium-intensity movement are deliberately absent at this stage, reflecting an incremental acquisition strategy that prioritises signal quality and annotation reliability over immediate coverage of every target activity. For the activity-recognition experiments reported in the following sections, the usable portion of the dataset comprises 110 labelled segments across the seven classes with sufficient sample counts for supervised training: eating with cutlery, laying, seating, showering, standing, walking and washing teeth.

2.2. Behavioural Monitoring Framework

Detecting routine deviation in principle requires continuous, longitudinal behavioural sequences. In practice, the wearable recordings described in Section 2.1 exist only as isolated start-stop activity segments, not continuous multi-hour or multi-day streams. The framework bridges this gap with a hybrid strategy, real-world data is used to validate activity recognition, and the resulting activity vocabulary then seeds a synthetic routine generator that produces the continuous sequences required for routine-level learning and anomaly detection.

A daily routine is represented as an ordered sequence $X = \{x_1, x_2, \dots, x_T\}$, where x_t denotes the activity occupying time step t . Each step corresponds to a 5-minute interval, giving $T = 288$ steps per day. The detection task is

to estimate, for every step, whether the observed activity is consistent with the individual's learned behavioural model, not merely whether the activity itself is rare, but whether it is anomalous given its timing, duration and surrounding context. The framework is therefore a monitoring and decision-support tool rather than a diagnostic one: it does not determine whether a person has dementia, or of which type, but flags deviations from that person's expected routine for review by caregivers and clinicians.

Activity recognition serves as a supporting layer that converts multimodal wearable signals into the categorical labels consumed by routine modelling. Five classical classifiers were evaluated for this role: Logistic Regression, k-Nearest Neighbours, Random Forest, XGBoost and LightGBM, chosen for their suitability to a dataset of modest size, with results reported in Section 3. Because the anomaly detection stage reasons over long-term temporal structure rather than isolated labels, occasional confusion between semantically close activities at this layer is not treated as a critical failure mode.

The two stages are connected through the activity vocabulary rather than through the raw signals. The anomaly detector consumes sequences of categorical activity labels at 5-minute resolution; in the target system, these labels are produced by the activity-recognition layer, whose window-level predictions are aggregated into one label per step.

Synthetic routines are constructed by dividing each 24-hour day into five time blocks grounded in dementia-care observational literature, Figure 3: morning (07:00–10:00), late morning (10:00–12:00), afternoon (12:00–18:00), evening (18:00–22:00) and night (22:00–07:00), each associated with a characteristic activity profile. Within each block, activities are drawn probabilistically from five behavioural categories: hygiene, meals, sleep and rest, mobility, and sedentary behaviour, so that, for example, sleep-related states dominate the night block while meals, hygiene and light mobility are concentrated in daytime and evening blocks.

Each category is further assigned a typical duration range in time steps (hygiene activities span roughly 10–30 minutes and meals 15–45 minutes, for instance) with randomised shifts in timing, ordering and duration introduced to capture realistic inter-individual and day-to-day variability.

Anomalies are injected into otherwise normal routines through three mechanisms chosen for clinical interpretability rather than random noise, drawing on documented dementia-related behavioural disturbance. Context-inappropriate insertion places an activity in a time block where it is behaviourally implausible, such as high-intensity mobility at night or prolonged sleep in the afternoon, simulating patterns such as nocturnal wandering or irregular sleep cycles.

Activity omission removes an expected element of the routine, such as a meal or a hygiene activity, reflecting scenarios like reduced appetite or self-care neglect.

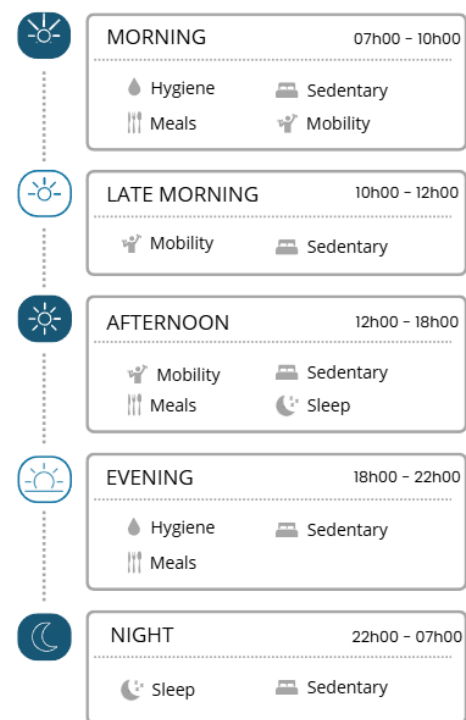


Figure 3. Time-of-day blocks used for synthetic daily routine generation, with the characteristic activity categories associated with each period

Duration and sequence alteration extends or shortens an activity beyond its typical range, or reorders activities in a way that breaks the expected behavioural flow. A fixed share of steps in each synthetic sequence is designated anomalous, keeping the resulting deviations rare relative to normal behaviour, as they would be in a real deployment.

Each 288-step routine is first mapped from categorical activity labels to integer indices and passed through a learnable embedding of dimension 8, allowing the model to capture latent semantic relationships between activity types rather than treating them as unrelated symbols. The embedded sequence is processed by two stacked Bidirectional Gated Recurrent Unit (BiGRU) layers with 128 hidden units per direction. The GRU's update and reset gates let the model retain stable behavioural patterns while remaining responsive to abrupt change, and the bidirectional configuration lets every time step draw on both past and future context within the same day, a property well suited to detecting anomalies that manifest as unexpected ordering or atypical duration rather than an unusual activity in isolation. A final fully connected layer with a sigmoid activation produces a per-step anomaly probability, with a threshold at 0.50 to yield a binary label. The model is trained with the AdamW optimiser (learning rate 3×10^{-4} , batch size 256) under binary cross-entropy loss, with early stopping on validation loss to control overfitting.

3. Results

This section reports the experimental evaluation of the proposed pipeline. Consistent with the two-stage design described in Section 2, the analysis is organised in two parts: Section 3.1 evaluates activity recognition on the real-world wearable dataset, validating the mapping from multimodal sensor signals to meaningful behavioural states; Section 3.2 evaluates the sequential anomaly detection model on synthetic daily routines with injected behavioural deviations, assessing its ability to learn routine structure and identify context-sensitive anomalies over time.

3.1. Real-Data Validation: Activity Recognition

The five classifiers were trained and evaluated under an identical protocol on the 110 labelled real segments spanning seven activity classes. Table 2 reports overall performance; LightGBM achieved the best results, with XGBoost and Random Forest close behind, while kNN trailed substantially.

Table 3 breaks down the best-performing model's results by class. Recognition is strongest for activities with structured, repetitive motion, eating with cutlery and washing teeth, while purely postural classes are harder to separate from wrist-worn signals alone; standing, in particular, was not correctly recognised in this evaluation.

3.2. Synthetic-Data Validation: Anomaly Detection

The anomaly detection model was trained in a semi-supervised setting on 8,000 normal synthetic daily sequences and evaluated on a held-out set of 20,000 sequences, of which 10,000 contained injected anomalies affecting roughly 15% of their time steps. Restricting anomalies to the evaluation phase allowed the model's generalisation to previously unseen deviations to be assessed directly. Under this protocol, the model reached an overall step-level accuracy of 99.4%, with an AUROC of 0.999 and a PR-AUC of 0.989. Table 4 reports precision, recall and F1-score separately for each class.

Table 2. Overall activity recognition performance, ordered by decreasing accuracy

Model	Accuracy	Macro Precision	Macro Recall	Macro F1 Score	Weighted F1 Score
LightGBM	0.73	0.64	0.67	0.64	0.70
XGBoost	0.70	0.61	0.65	0.62	0.66
Random Forest	0.70	0.61	0.65	0.62	0.66
Logistic Regression	0.59	0.58	0.59	0.55	0.55
kNN	0.41	0.35	0.41	0.32	0.29

Table 3. Per-class performance of the best-performing model (LightGBM), ordered by decreasing precision

Activity	Precision	Recall	F1-score
Eating with cutlery	1.00	0.78	0.88
Washing teeth	0.88	1.00	0.93
Laying	0.80	0.57	0.67
Showering	0.67	0.67	0.67
Seating	0.58	0.88	0.70
Walking	0.56	0.83	0.67
Standing	0.00	0.00	0.00

Table 4. Precision, recall and F1-score for normal and anomalous routine steps

Class	Precision	Recall	F1-score
Normal	0.995	0.998	0.997
Anomalous	0.978	0.941	0.959

False positives were not uniformly distributed across activity categories. Hygiene-related steps accounted for the largest share (665 occurrences), followed by lying (304), meals (127) and sleep (116); mobility-related categories were far less affected, with medium movement, light movement, walking and standing contributing 59, 39, 26 and 1 false positives respectively. This pattern is consistent with the naturally flexible timing and duration of hygiene, rest and meal activities, which legitimately vary even under normal conditions and sit closer to the model's decision boundary near time-block transitions.

4. Discussion

Across both stages of the pipeline, the results support the technical feasibility of the proposed approach while pointing to clear next steps before clinical translation. On activity recognition, ensemble methods (LightGBM, XGBoost, Random Forest) consistently outperformed simpler baselines, in line with the wider HAR literature [19], [20], and structured, repetitive activities such as walking and washing teeth were recognised more reliably than static postures, which are inherently harder to separate using wrist-mounted sensors alone. Because the anomaly detection stage reasons over routine structure rather than isolated labels, it is expected to be partly tolerant to such errors, although this has not yet been quantified.

The synthetic routine generator is arguably the more consequential contribution here, precisely because it sidesteps the central obstacle identified in Section 1, the scarcity of long-term, labelled behavioural data from people living with dementia. By encoding empirically grounded time-of-day structure, activity duration ranges and controlled variability, it enables large-scale, systematically annotated experimentation that would otherwise be infeasible with real-world data alone. The anomaly detection model's strong separability (AUROC 0.999, PR-AUC 0.989) demonstrates that the BiGRU architecture can learn this structure effectively and detect the three classes of injected deviation. These figures should nonetheless be read as an upper bound: performance on purely synthetic, controlled sequences is likely to be optimistic relative to noisy, less predictable real-world data, and the concentration of false positives around hygiene, rest and meal activities, the categories with the most natural timing variability, hints at where that optimism would be tested first.

A related concern is whether the detector learns clinically meaningful deviations or merely the rules of the generator. Because normal routines and injected anomalies derive from the same assumptions (time-of-day blocks, category-specific durations and plausible orderings), the reported separability shows that the BiGRU recovers the structure encoded by the generator, not that it would detect real behavioural change in people living with dementia. Excluding anomalies from training and defining deviations relative to context mitigate, but do not remove, this risk. Evaluation on unseen anomaly types, alternative generator settings and, ultimately, real longitudinal recordings is therefore required before any claim about clinical sensitivity can be made.

A further issue is that the anomaly detector does not yet operate on wearable signals: the real recordings establish the activity vocabulary and evaluate activity recognition, whereas the detector is trained and tested on synthetic sequences that assume error-free labels. Recognition errors (macro F1-score of 0.64, with standing not recognised) could appear to the detector as spurious context violations or mask genuine ones, and part of the routine vocabulary, such as sleep and medium-intensity movement, still lacks a validated sensor-level counterpart. The next step is to inject recognition errors sampled from the empirical confusion matrix into the synthetic sequences and, as continuous recordings become available, to evaluate the detector on sequences produced directly by the recognition stage.

Several limitations temper these results. The real-world dataset, while adequate for validating the acquisition pipeline, is small (110 usable segments), drawn from healthy young adults rather than the target clinical population, and does not yet reflect the motor or cognitive variability associated with dementia. Anomaly detection was evaluated exclusively on synthetic data; no comparison was made against alternative anomaly detectors, from probabilistic sequence models (Markov chains, HMMs) and one-class methods (Isolation Forest, One-Class SVM) to transformer- and autoencoder-based models, which remain necessary reference points for future benchmarking. Moving from this proof of concept to a deployable system would also require addressing practical constraints largely outside the present scope: device adherence and comfort for people with dementia, graceful handling of missing data during sensor-off periods, latency budgets suited to real-time caregiver alerts, model drift as routines evolve with disease progression (calling for periodic recalibration or continual learning), and the risk of alert fatigue, which argues for interpretable, actionable outputs rather than raw anomaly scores.

The synthetic generator has not yet been validated quantitatively against real routines. Although its time-of-day blocks and duration ranges are grounded in dementia-care literature, showing that generated routines reproduce properties of real daily behaviour (activity-duration distributions, hourly activity profiles and transition structure), against time-use data for older adults or longitudinal project recordings, remains a prerequisite for

interpreting detection results beyond the synthetic setting. Nor has the framework been validated against clinical information. Its intended role is to flag deviations from an individual's routine for clinical follow-up, not to diagnose dementia or its subtypes; clinical validation will require larger, multi-site and geographically diverse cohorts of people living with dementia, characterised with demographic, medical, medication, neuropsychological and functional data.

Overall, the system should be regarded as a proof of concept at an early technology readiness level. These limitations do not invalidate the conceptual approach, but they define the evidence still required and reinforce the case for a hybrid strategy that combines real acquisition data, synthetic augmentation and interpretable models, iteratively refined as more longitudinal real-world data becomes available.

5. Conclusion and Future Work

This article reported the joint development of a real-world wearable dataset for human activity recognition and a routine-aware behavioural monitoring framework for dementia care, describing how the two are designed to work together within a single project. The dataset validates a dual-device acquisition and multimodal alignment pipeline across ten short-duration daily activities; the monitoring framework shows how the resulting activity vocabulary can be combined with knowledge-informed synthetic routine generation and a BiGRU-based sequential model to detect context-dependent behavioural deviations with high fidelity under controlled conditions. The framework bridges this gap with a hybrid strategy, extending the routine-generation approach

Future work will extend the real-world dataset toward longer recordings, a broader activity repertoire, and participants representative of the target population, ideally including individuals with cognitive impairment recorded in naturalistic, free-living settings as the custom wearable hardware matures. On the modelling side, priorities include benchmarking the anomaly detection model against transformer- and autoencoder-based baselines, exploring personalised routine models that adapt to individual behavioural drift over time, and evaluating alternative multimodal synchronisation strategies. Together, these steps are intended to move the framework from a validated proof of concept toward a system suitable for evaluation in real dementia-care environments.

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